

Class 13: RNASeq Analysis

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Background

Today we will analyze some RNA sequencing data on the effects of a common steroid drug on airway cell lines.

There are two main inputs we need for their analysis:

- countData: counts for genes in rows with experiments in the columns
- colData: or metadata

Import countData and colData

```
counts <- read.csv("airway_scaledcounts.csv", row.names=1)
metadata <- read.csv("airway_metadata.csv")
```

Q1. How many genes are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

Q2. How many ‘control’ cell lines do we have?

```
sum(metadata$dex == "control")
```

```
[1] 4
```

Toy Differential Gene Expression

Let’s try finding the average or mean of the “control” and “treated” columns and see if they differ.

- First we need to find all “control” columns
- Extract just the “control” values for each gene
- Calculate the `mean()` for each gene “control” values

```
metadata
```

	id	dex	celltype	geo_id
1	SRR1039508	control	N61311	GSM1275862
2	SRR1039509	treated	N61311	GSM1275863
3	SRR1039512	control	N052611	GSM1275866
4	SRR1039513	treated	N052611	GSM1275867
5	SRR1039516	control	N080611	GSM1275870
6	SRR1039517	treated	N080611	GSM1275871
7	SRR1039520	control	N061011	GSM1275874
8	SRR1039521	treated	N061011	GSM1275875

```
all(colnames(counts) == metadata$id)
```

```
[1] TRUE
```

the dex column tells me whether we have “control” or “treated”

```
control.inds <- metadata$dex == "control"
```

Extract just the “control” values for all genes

```
control.counts <- counts[,control.inds]
```

Calculate the mean value for each gene in these control columns

```
control.mean <- rowMeans( control.counts )  
head(control.mean)
```

```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460  
          900.75           0.00           520.50           339.75           97.25  
ENSG00000000938  
          0.75
```

Q3. Do the same for the “treated” values to get a `treated.mean`

```
treated.inds <- metadata$dex == "treated"  
treated.counts <- counts[,treated.inds]  
treated.mean <- rowMeans( treated.counts )  
head(treated.mean)
```

```
ENSG00000000003 ENSG00000000005 ENSG00000000419 ENSG00000000457 ENSG00000000460  
          658.00           0.00           546.00           316.50           78.75  
ENSG00000000938  
          0.00
```

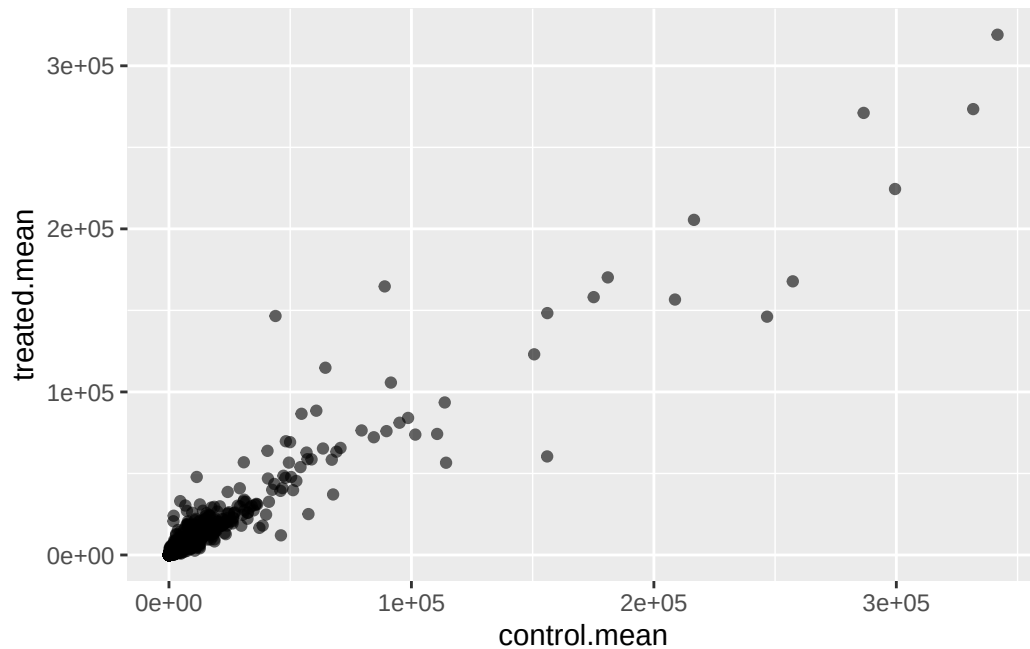
Q4. Make a plot of `control.mean` vs `treated.mean`

Lets store our mean values together in a data.frame for easier book-keeping.

```
meancounts <- data.frame(control.mean, treated.mean)  
head(meancounts)
```

	control.mean	treated.mean
ENSG00000000003	900.75	658.00
ENSG00000000005	0.00	0.00
ENSG00000000419	520.50	546.00
ENSG00000000457	339.75	316.50
ENSG00000000460	97.25	78.75
ENSG00000000938	0.75	0.00

```
library(ggplot2)
ggplot(meancounts) +
  aes(control.mean, treated.mean) +
  geom_point(alpha=.6)
```

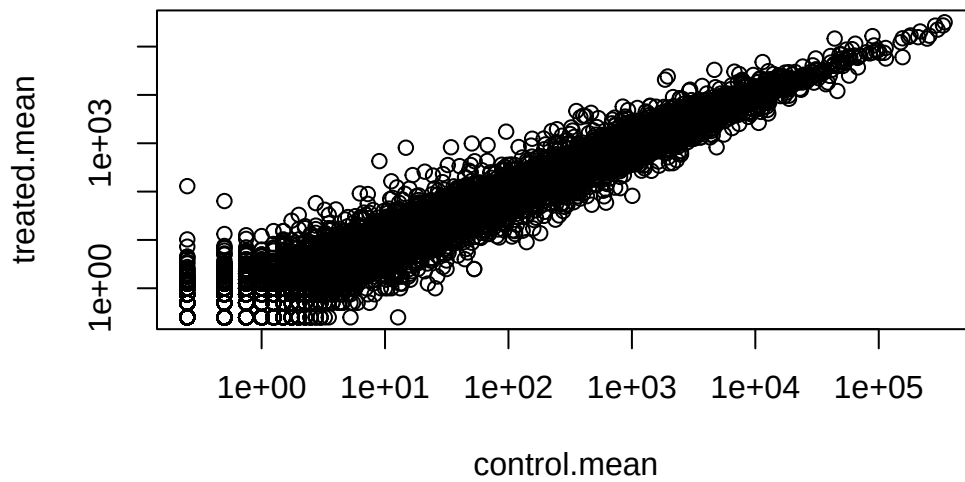


We totally need to log transform this data as it is heavily skewed!

```
plot(meancounts, log="xy")
```

Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted from logarithmic plot

Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted from logarithmic plot



Amount is the same

```
log(20/20)
```

```
[1] 0
```

Doubling the amount

```
log(40/20)
```

```
[1] 0.6931472
```

Halving the amount

```
log(10/20)
```

```
[1] -0.6931472
```

A common rule of thumb is to focus on genes with a log₂ “fold change” of +2 as so called UP REGULATED and -2 as DOWN REGULATED.

```
log2(80/20)
```

```
[1] 2
```

Lets add a log2 fold change value to our `meancounts` dataframe.

```
meancounts$log2fc <-log2(meancounts$treated.mean/meancounts$control.mean)
head(meancounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000005	0.00	0.00	NaN
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000938	0.75	0.00	-Inf

Q. Remove any “zero count” genes from the dataset for further analysis

```
to.keep <- rowSums(meancounts[,1:2] == 0) == 0
sum(to.keep)
```

```
[1] 21817
```

```
mycounts <- meancounts[to.keep,]
head(mycounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000971	5219.00	6687.50	0.35769358
ENSG000000001036	2327.00	1785.75	-0.38194109

Q. How many genes are “up” regulated a a log2fc threshold of +2?

```
sum(mycounts$log2fc >= 2)
```

[1] 314

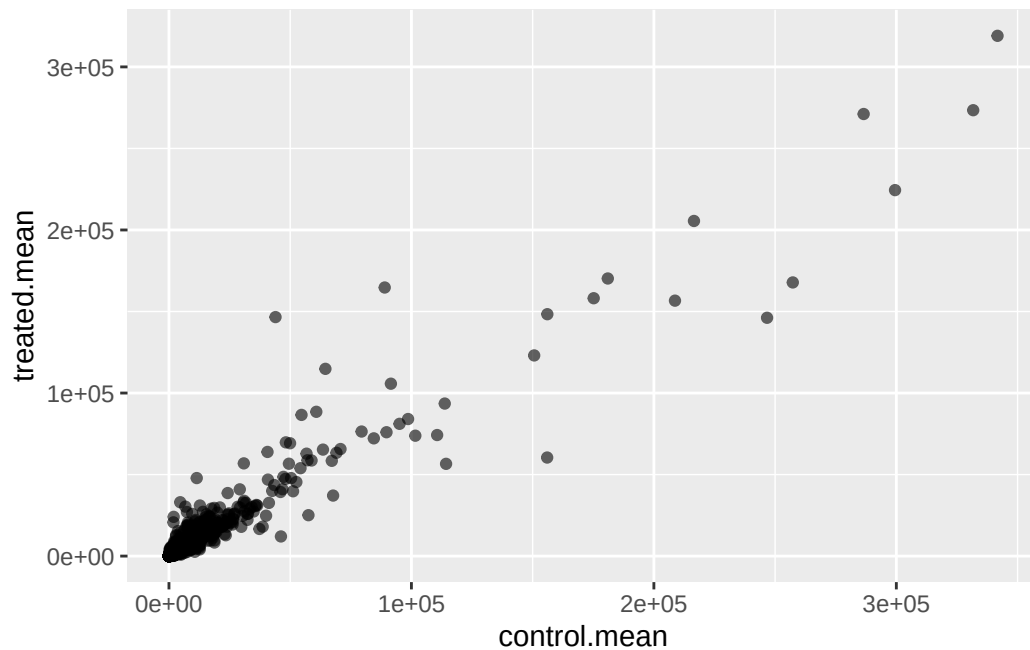
Q. How many genes are “down” regulated a a log2fc threshold of -2?

```
sum(mycounts$log2fc <= -2)
```

[1] 485

However, there are some stats missing from this analysis, such as how significant these changes are and/or whether differences between the control and treated may have other causes.

```
ggplot(mycounts) +  
  aes(control.mean, treated.mean) +  
  geom_point(alpha=.6)
```



Setting up for DESeq

We will use DESeq2 to do this:

```
library(DESeq2)
```

The first function we will use from this package sets up the input in the particular format that DESeq wants.

```
dds <- DESeqDataSetFromMatrix(countData = counts,  
                              colData = metadata,  
                              design = ~dex)
```

converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

We can now run our DESeq analysis

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds)
```

Peek at results

```
head(res)
```


log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

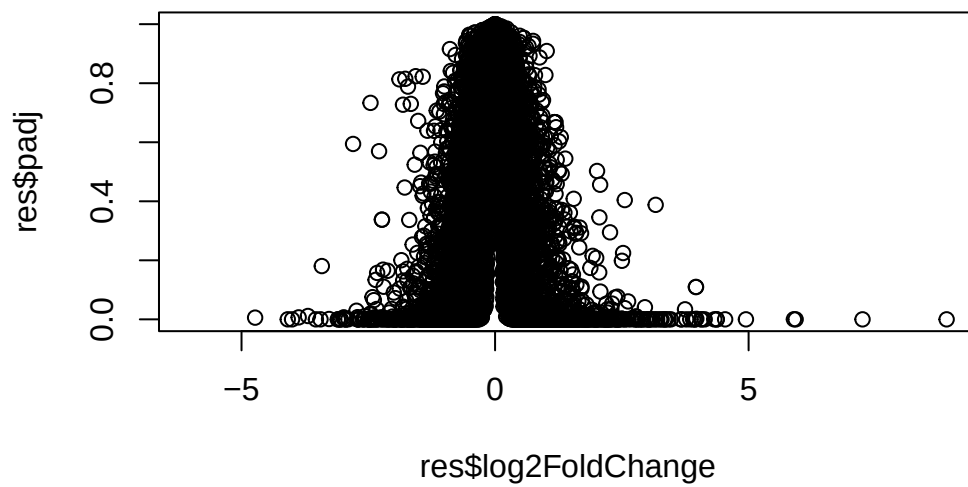
DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG0000000000419	520.134160	0.2061078	0.101059	2.039475	0.0414026
ENSG0000000000457	322.664844	0.0245269	0.145145	0.168982	0.8658106
ENSG0000000000460	87.682625	-0.1471420	0.257007	-0.572521	0.5669691
ENSG0000000000938	0.319167	-1.7322890	3.493601	-0.495846	0.6200029
	padj				
	<numeric>				
ENSG0000000000003	0.163035				
ENSG0000000000005	NA				
ENSG0000000000419	0.176032				
ENSG0000000000457	0.961694				
ENSG0000000000460	0.815849				
ENSG0000000000938	NA				

Result figure: Volcano Plot

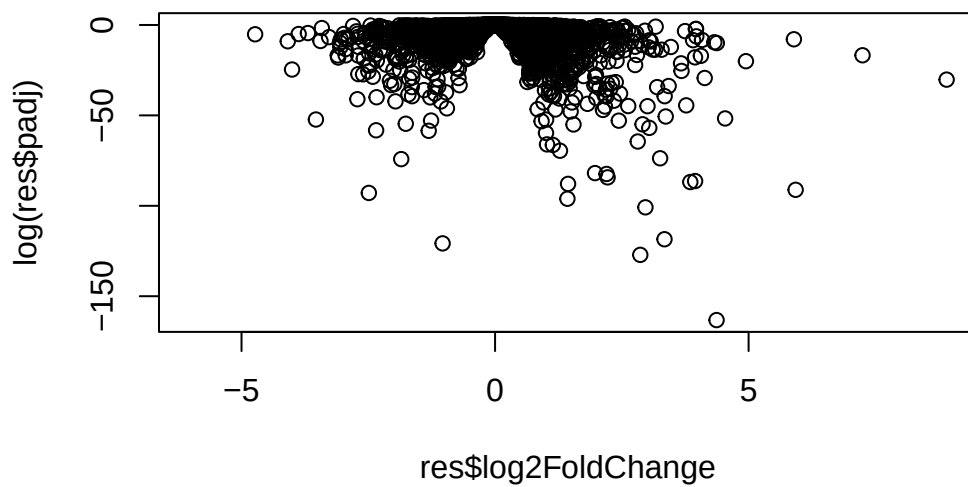
plot of the P values the Log2FC

```
plot(res$log2FoldChange, res$padj)
```



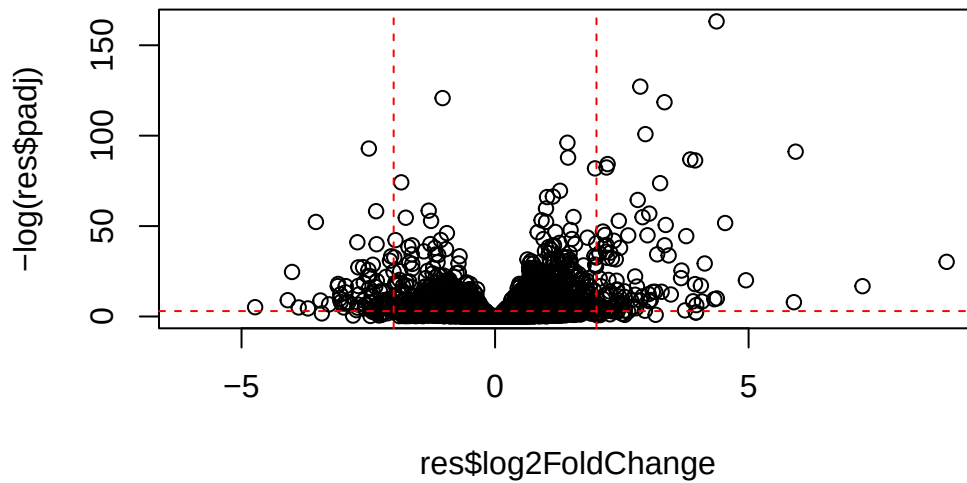
This P-value is again heavily skewed so lets log transform it.

```
plot(res$log2FoldChange, log(res$padj))
```

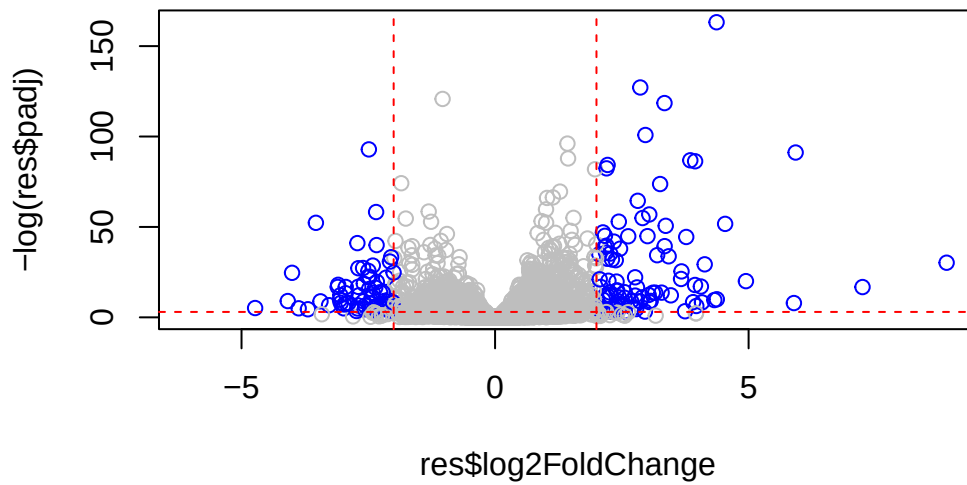


We can flip the y-axis by adding a minus sign. This will make it easier to interpret.

```
plot(res$log2FoldChange, -log(res$padj))  
# Add some cut-off lines  
abline(v=c(-2,2), col="red", lty=2)  
abline(h=-log(0.05), col="red", lty=2)
```



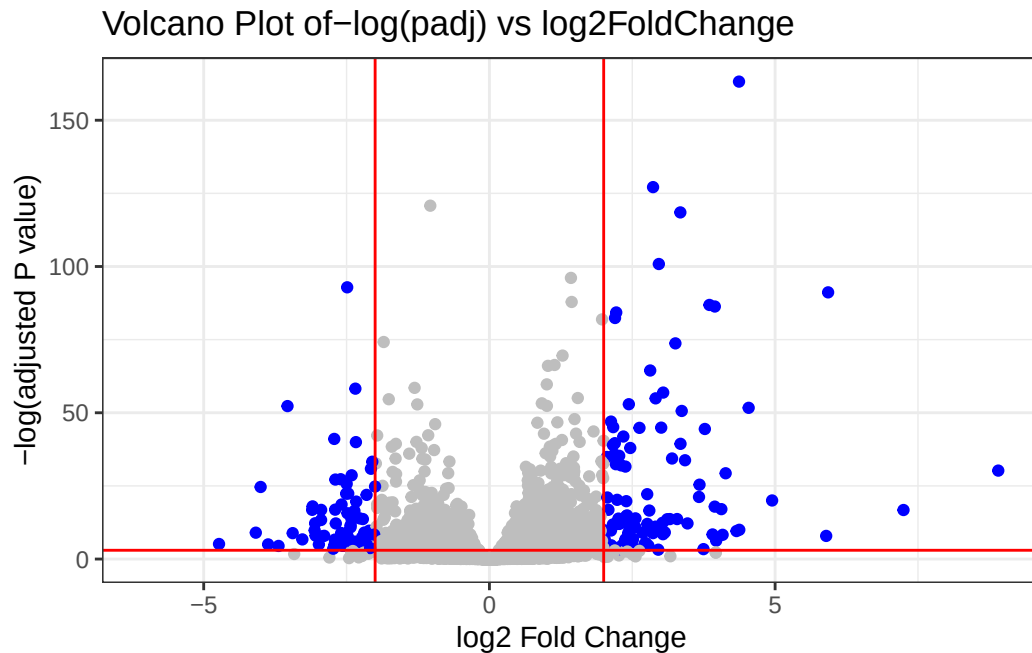
```
mycols <- rep("grey", (nrow(res)))  
mycols[ abs(res$log2FoldChange) >= 2 ] <- "blue"  
mycols[ res$padj > 0.05 ] <- "grey"  
plot(res$log2FoldChange, -log(res$padj), col=mycols)  
  
abline(v=c(-2,2), col="red", lty=2)  
abline(h=-log(0.05), col="red", lty=2)
```



Q. Make a ggplot volcano plot with colors and lines as annotation along with nice axis labels.

```
ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange, y = -log(padj)) +
  geom_point(col = mycols) +
  labs(title = "Volcano Plot of -log(padj) vs log2FoldChange",
       x = "log2 Fold Change",
       y = "-log(adjusted P value)") +
  geom_vline(xintercept = c(-2, 2), color = "red") +
  geom_hline(yintercept = -log(0.05), color = "red") +
  theme_bw()
```

Warning: Removed 23549 rows containing missing values or values outside the scale range (`geom\_point()`).



We need to add gene annotation

Gene symbols and different database IDs

Pathway Analysis

Find what biological pathways my differentially expressed genes participate in.

We first need to add gene symbols (e.g. HBB, etc.) so we know what genes we are dealing with. We need to “translate” between ENSEMBLE ids that we have in the rownames of **res**.

```
head(rownames(res))
```

```
[1] "ENSG000000000003" "ENSG000000000005" "ENSG000000000419" "ENSG000000000457"
[5] "ENSG000000000460" "ENSG000000000938"
```

```
#Install from bioconductor with `BiocManager::install("`
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
columns(org.Hs.eg.db)
```

[1]	"ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6]	"ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11]	"GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16]	"OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21]	"PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26]	"UNIPROT"				

```
res$symbol <- mapIds(org.Hs.eg.db,  
  keys=row.names(res), # Our genenames  
  keytype="ENSEMBL",   # The format of our genenames  
  column="SYMBOL",     # The new format we want to add  
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$genename <- mapIds(org.Hs.eg.db,  
  keys=row.names(res),  
  keytype="ENSEMBL",  
  column = "GENENAME")
```

'select()' returned 1:many mapping between keys and columns

```
res$uniprot <- mapIds(org.Hs.eg.db,  
  keys=row.names(res),  
  keytype="ENSEMBL",  
  column = "UNIPROT")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(org.Hs.eg.db,  
  keys=row.names(res),  
  keytype="ENSEMBL",  
  column = "ENTREZID")
```

'select()' returned 1:many mapping between keys and columns

Be sure to save our annotated results to a file.

```
write.csv(res,file="my_annotated_results.csv")
```

```
library(gage)
library(gageData)
library(pathview)
```

```
data(kegg.sets.hs)
```

```
# Examine the first 2 pathways in this kegg set for humans
head(kegg.sets.hs, 2)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

To run pathway analysis, we will use the `gage()` function and it requires a wee “vector of importance”. We will use our Log2FC results from our `res` object.

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
      7105      64102      8813      57147      55732      2268
-0.35070302      NA  0.20610777  0.02452695 -0.14714205 -1.73228897
```

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

What is in the returned `keggres` object?

```
attributes(keggres)
```

```
$names  
[1] "greater" "less"    "stats"
```

```
# Look at the first three down (less) pathways  
head(keggres$less, 3)
```

	p.geomean	stat.mean	p.val
hsa05332 Graft-versus-host disease	0.0004250461	-3.473346	0.0004250461
hsa04940 Type I diabetes mellitus	0.0017820293	-3.002352	0.0017820293
hsa05310 Asthma	0.0020045888	-3.009050	0.0020045888

	q.val	set.size	exp1
hsa05332 Graft-versus-host disease	0.09053483	40	0.0004250461
hsa04940 Type I diabetes mellitus	0.14232581	42	0.0017820293
hsa05310 Asthma	0.14232581	29	0.0020045888

We can pass our foldchanges vector (our results) together with any of these highlighted pathway IDS to see how our genes overlap the pathway.

```
pathview(gene.data=foldchanges, pathway.id="hsa05310")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/leah/Documents/UCSD/BIMM 143/class13
```

```
Info: Writing image file hsa05310.pathview.png
```

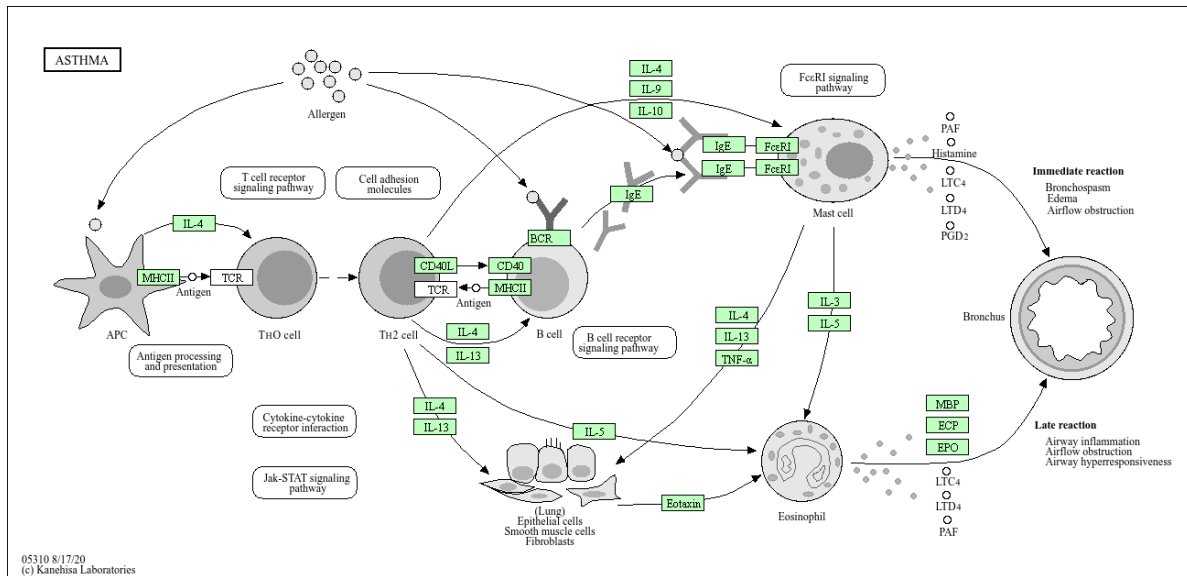



Figure 1: The asthma pathway overlaps with our differentially expressed genes